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Driving with LLMs: Fusing Object-Level Vector Modality for Explainable Autonomous Driving

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***Abstract*— Large Language Models (LLMs) have shown promise in the autonomous driving sector, particularly in generalization and interpretability. We introduce a unique object-level multimodal LLM architecture that merges vectorized numeric modalities with a pre-trained LLM to improve context understanding in driving situations. We also present a new dataset of 160k QA pairs derived from 10k driving scenarios, paired with high quality control commands collected with RL agent and question answer pairs generated by teacher LLM (GPT-3.5). A distinct pretraining strategy is devised to align numeric vector modalities with static LLM representations using vector captioning language data. We also introduce an evaluation metric for Driving QA and demonstrate our LLM-driver's proficiency in interpreting driving scenarios, answering questions, and decision-making. Our findings highlight the potential of LLM-based driving action generation in comparison to traditional behavioral cloning. We make our benchmark, datasets, and model available[1] for further exploration.**

I. Introduction

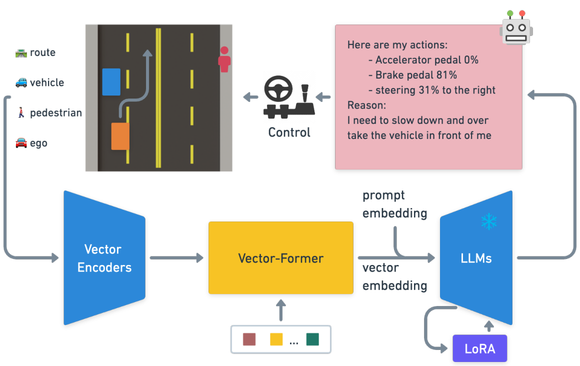

Fig. 1: An overview of the architecture for Driving with LLMs, demonstrating how object-level vector input from our driving simulator is employed to predict actions via LLMs

Remarkable abilities of Large Language Models (LLMs) demonstrate early signs of artificial general intelligence (AGI) [1], exhibiting capabilities such as out-of-distribution (OOD) reasoning, common sense understanding, knowledge retrieval, and the ability to naturally communicate these aspects with humans. These capabilities align well with the focus areas of autonomous driving and robotics in general [2] [3].

Modern scalable autonomous driving systems, whether they adopt an end-to-end approach using a single network [4], or a component-based configuration that combines learnable perception and motion planning modules [5] [6], face common challenges. These systems often behave as 'black-boxes' in the decision making process, making it especially difficult to endow them with OOD reasoning and interpretability capabilities. Such issues persist even though there have been some strides towards addressing them [7].

Textual or symbolic modality, with its inherent suitability for logical reasoning, knowledge retrieval, and human communication, serves as an excellent medium for harnessing the capabilities of LLMs [8]. However, its linear sequential nature poses limitations for nuanced spatial understanding [1], a crucial aspect of autonomous navigation. Pioneering work in Visual Language Models (VLMs) has begun to bridge this gap by merging visual and text modalities [9], enabling spatial reasoning with the power of pre-trained LLMs. However, to effectively incorporate the new modality into the language representation space requires extensive pretraining with a significant volume of labeled image data.

We propose a novel methodology for integrating numeric vector modality, a type of data frequently used in robotics for representing speed, actuator positions and distance measurements, into pre-trained LLMs. Such modality is considerably more compact than vision alleviating some of the VLM scaling challenges. Specifically, we fuse vectorized object-level 2D scene representation, commonly used in autonomous driving, into a pre-trained LLM with adapters [10]. This fusion enables the model to directly interpret and reason about comprehensive driving situations. As a result, the LLMs are empowered to serve as the "brain" of the autonomous driving system, interacting directly with the simulator to facilitate reasoning and action prediction.

To obtain training data in a scalable way, we first use a custom 2D simulator and train a reinforcement learning (RL) agent to solve the driving scenarios and serve in place of a human driving expert. To ground the object-level vector into the LLMs, we introduce a language generator that translates this numerical data into textual descriptions for representation pretraining. We further leverage a teacher LLM (GPT) to generate question answering dataset conditioned on the language descriptions of 10k different driving scenarios. Our model firstly undergoes a pretraining phase that enhances alignment between the numeric vector modality and the language latent representations. Following this, we train our unique architectural design to establish a robust baseline model LLM-driver for the driving action prediction and driving question answering tasks. We provide our datasets, evaluation benchmarks and a pre-trained model[1] for reproducibility and

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[1]https://github.com/wayveai/Driving-with-LLMs

hope to inspire and facilitate further advancements in the field. The subsequent sections of this paper detail the theoretical background, our proposed architecture and experimental setup, our preliminary results, potential directions for future research, and the implications of our work for the broader field of autonomous driving.

In this paper, we have made the following contributions:

1) **Novel object-level multimodal LLM architecture:** We propose a novel architecture that fuses an object-level vectorized numeric modality into any LLMs with a two-stage pretraining and finetuning method.
2) **Driving scenario QA task and a dataset:** We provide a 160k question-answer pairs dataset on 10k driving situations with control commands, collected with RL expert driving agents and an expert LLM-based question answer generator. We also provide the whole methodology for collecting more data.
3) **Novel Driving QA (DQA) evaluation and a pre-trained baseline:** We present a novel way to evaluate Driving QA performance using the same expert LLM grader. We provide initial evaluation results and a baseline using our end-to-end multimodal architecture.

Our work provides the first-of-its-kind baseline approach for integrating LLMs into driving task in simulation. This includes a comprehensive framework encompassing the simulator, automatic data collection, the integrating of a new object-level vector modality into LLMs, and the GPT-based evaluation approaches. We are excited about the potential of these advancements to revolutionize the autonomous driving landscape and look forward to seeing the transformative research directions they will bring.

II. Related Works

A. End-to-End Autonomous Driving Systems

There was a large progress in end-to-end deep learning methods for autonomous systems in recent years [11], [12], [4]. With some of the earliest efforts dating back to ALVINN [13] and more recent works such as [14]. However, a fundamental challenge with modern autonomous driving systems is the lack of interpretability in the decision making process [15]. Understanding why a decision is made is crucial for understanding areas of uncertainty, building trust, enabling effective human-AI collaboration and ensuring safety [16]. We continue this line of research and add vector/textual modality and pretrained LLMs to the end-to-end autonomous driving.

B. Interpretability of Autonomous Driving Systems

A variety of explainability methods have been introduced [17] in order to understand the underlying decision process of deep neural networks. For example [18], [19] and [20] are well-established model-agnostic interpretability methods that generate explanations for individual predictions. Other methods such as gradient based [21], saliency maps [22] and attention maps [23] target the inner operations of models to explain the decision making process. In the field of autonomous vehicles, visual attention maps which highlight causally influential regions in driving images were proposed in [24]. In [25] the authors combined attention based methods with natural language into an attention-based vehicle controller to provide natural language action descriptions and explanations based on a series of image frames. This work was further extended in [26], where the authors improved the architecture by integrating part of speech prediction and special token penalties. Others argue that attention is not enough [27], and so multiple efforts have been made trying to combine this with other explanatory methods. For example [28] propose to explain transformers by leveraging attentive class activation tokens, encoded features, their gradients, and their attention weights all together. Building on this research, we are proposing to use text modality for explainability in autonomous driving.

C. Multi-modal LLMs in Driving Tasks

In recent times, there has been a notable trend towards integrating multiple modalities into unified large-scale models. Notable examples include VLMs such as [29], [30], [31], and [32], which effectively combine language and images to accomplish tasks like image captioning, visual question answering, and image-text similarity. Another noteworthy advancement [33] involves the fusion of information from six distinct modalities: text, image/video, audio, depth, thermal, and inertial measurements. This exciting development expands the possibilities for generating content using diverse data input and output types, while also enabling broader multi-modal search capabilities.

With camera sensors being one of the most common sensors used in autonomous driving [34], a natural step to incorporate language has been through VLMs. For example [35] use images and language directions to train a driving policy. [36] propose a method for learning vehicle control with the help of human advice. The system learns to summarize its visual observations in natural language, predict an appropriate action response (e.g. “I see a pedestrian crossing, so I stop”), and predict the controls, accordingly. Using language to explain the inner workings of the model has also been used in [37], where user-friendly natural language narrations and reasoning are provided for each decision making step of autonomous vehicular control and action. In robotics, we have seen efforts fusing language with other modalities. Albeit outside of autonomous driving field, the closest work to ours [38] utilises point clouds with 3D bounding boxes of potential object candidates, and a language utterance referring to a target object in the scene, to train a model capable of identifying a target object from a set of potential candidates. Recently, the RT-2 paper [39] demonstrated a similar approach in utilizing LLMs for low-level robotics control tasks, including the joint training of VQA and control tasks. However, their framework is confined to the vision modality, whereas we introduce a novel methodology for grounding vector-based object level modalities into LLMs, facilitating interpretable control and driving QA tasks. In contrast to these efforts, the work presented in this paper, to the best of our knowledge, is the first to fuse numeric vector modality with language in the domain of autonomous vehicles.

III. METHOD

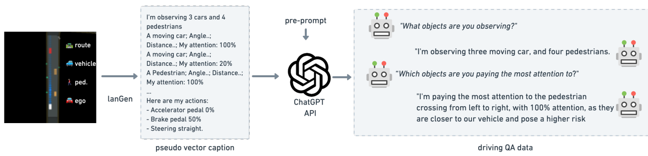

Fig. 2: The illustration of our Driving QA Dataset automatic labelling process

A. Data Collection using RL experts

To generate language-based grounded driving datasets, we use a custom-built realistic 2D simulator with procedural generation of driving scenarios. We use RL agent that solves the simulated scenarios using an object-level ground-truth representation of the driving scene. In our approach, we map a vector representation of the environment to an action for the vehicle dynamics with an attention-based neural network architecture. This model was optimized using Proximal Policy Optimization (PPO) [40]. Subsequently, we collect continuous driving data from 15 diverse virtual environments with randomly generated traffic conditions. Our collection includes a 100k dataset for pretraining, a 10k set for QA labeling and fine-tuning, and a 1k set dedicated to evaluation.

B. Structured Language Generation for Pseudo Vector Captioning

In our framework, we aim to convert vector representations into language using a structured language generator for the ease of grounding the vector representation into LLMs. Since our object-level vectors contain semantically significant attributes such as the number of cars and pedestrians, their respective locations, orientations, speeds, bounding boxes and other attributes, we employ a structured language generator (lanGen) function to craft pseudo language labels derived from the vector space, as illustrated below:

$\text{lanGen}(v_{\text{car}}, v_{\text{ped}}, v_{\text{ego}}, v_{\text{route}}, [o_{rl}]) \rightarrow$

"A moving car; Angle in degrees: 1.19; Distance: 9.98m; [My attention: 78%]
A pedestrian; Angle in degrees: -41.90; Distance: 11.94m; [My attention: 22%]
My current speed is 11.96 mph.
There is a traffic light and it is red. It is 12.63m ahead.
The next turn is 58 degrees right in 14.51m.

[Here are my actions:]
[- Accelerator pedal 0%]
[- Brake pedal 80%]
[- Steering straight]
"

In this function, variables v_{car}, v_{ped}, v_{ego}, and v_{route} denote vector information corresponding to cars, pedestrians, ego vehicle, and route, respectively. The optional term o_{rl} corresponds to the output from the RL agent, consisting of additional attention and action labels for guiding the action reasoning process. Attention labels are collected from RL policy attention layers similar to [41].

This lanGen enables the transformation of vector representations into human-readable language captions. It crafts a comprehensive narrative of the current driving scenario, inclusive of the agent's observations, the agent's current state, and its planned actions. This comprehensive contextual foundation enables the LLMs to conduct reasoning and construct appropriate responses in a manner that humans can interpret and understand.

The inclusion of o_{rl} is optional, and we generate two different versions of pseudo labels to cater to different requirements: 1) **Without Attention/Action:** Employed during the representation pre-training stage (see Subsection III-D.1), where the inference of attentions and actions is not required. 2) **With Attention/Action:** Utilized for VQA labeling with GPT during the fine-tuning stage (see Subsection III-D.2). This equips GPT with the capacity to ask specific questions about attentions and actions, empowering the driving LLM agent with the ability to reason about attentions and actions.

C. Driving QA Dataset Labeling

Large amount of data is the key to enabling the question answering ability of the language models. This becomes particularly crucial when a new modality is introduced to the LLMs; in such circumstances, it is essential to have a high-quality question-answering dataset relevant to the modality input. Studies have demonstrated that data labelled using ChatGPT surpasses crowd-sourced workers in terms of performance for text-annotation tasks [42]. Inspired from research such as Self-Instruct [43] and LLaVA [44], we utilize GPT to generate a Driving QA dataset. This dataset is conditioned on the structured language input, serving as a representative for the vectors.

To be specific, we tasked GPT to emulate the role of a certified professional driving instructor for generating questions and answers for students. We supplied basic driving rules as pre-prompts, covering aspects like traffic signal transitions and the driving side as we are operating in a left-handed driving country. We tasked GPT with creating 16 sets of random questions and answers per scenario, taking into account the current driving scenario provided by language description generated by the lanGen. We also include some instructions to guide the generation of action-related questions, so they can be easily parsed with regular expressions for evaluating action prediction tasks.

To reduce hallucination, we established simple guardrail rules. We instructed GPT to include a driving-related question that could not be inferred from the observation input. If faced with such a question, the model's response should be akin to "I'm unable to answer this question based on the observations I have," preventing the model from imagining objects not present in the provided vector observation.

Additionally, we included a random question unrelated to driving, instructing the model to respond with "As an AI Driver, the question you asked is out of my scope, but I can attempt to answer it." The model would then proceed to answer the question normally, thereby preserving the versatility of the language model as a general-purpose assistant.

Detailed instructions can be found in our open-sourced code[1]. Below are some examples of the question-answer pairs generated by GPT:

- **Question:** "What objects are you observing?" **Answer:** "I'm observing a moving car, and two pedestrians. The car is coming from the opposite direction, and the pedestrians are crossing the street from both the right and left sides."
- **Question:** "How are you going to drive in this situation and why?" **Answer:** "Here are my actions: - Accelerator pedal 0% - Brake pedal 7% - Steering straight. Reason: Because I'm maintaining a slow speed due to the pedestrians crossing, and I'm steering straight to keep our vehicle properly aligned with the road."
- **Question:** "What are the best tourist spots in London?" **Answer:** "As an AI Driver, the question you asked is out of my scope, but I can try to answer it. Some famous tourist spots in London include the Tower of London, Buckingham Palace, The British Museum, The Shard, and the London Eye."

D. Training the Driving LLM Agent

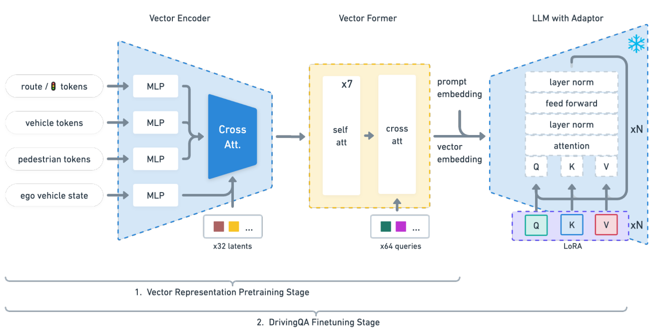

Fig. 3: The architecture of the Driving LLM Agent

Training the LLM-Driver involves formulating it as a Driving Question Answering (DQA) problem within the context of a language model. The key to this formulation is the integration of object-level vector modality with the pre-trained LLMs, creating a multi-modal system capable of interpreting and interacting with both language and vector inputs.

We use a two-stage process to train our model for effectively fusing the object-level vector modality into the LLM-Driver. In the first stage, we ground the vector representation into an embedding that can be decoded by the LLMs. This is done by freezing the language model and optimizing the weights of the vector encoders and the vector transformer. In the second stage, we finetune the model to the DQA task, training it to answer driving-related questions and take appropriate actions based on its current understanding of the environment.

As can be seen in the Figure 3, our model is built on three key components: the Vector Encoder, Vector Former, and a frozen LLM with a Low-Rank Adaptation (LoRA) [10] module.

- **Vector Encoder**: The four input vectors pass through the Multilayer Perceptron (MLP) layers. They're then processed by a cross-attention layer to move them into a latent space. We add the ego feature to each learned input latent to emphasize the ego states.
- **Vector Former**: This part contains self-attention layers and a cross-attention layer that work with the latent space and question tokens. This process transforms the latent vectors into an embedding that the LLM can decode.
- **LLM with Adaptor**: Here, we inject trainable rank decomposition matrices (LoRA) into the linear layers of the pretrained LLMs for parameter-efficient finetuning. We utilize LLaMA-7b [45] as the pretrained LLM for our experiments.

*1) **Vector Representation Pre-training**:* Integrating a new modality into pre-trained Large Language Models (LLMs) poses significant challenges due to the need for extensive data and computational resources. In this study, we propose a novel approach that uses structured language to bridge the vector space with the language embedding, with particular focus on numerical tokens.

During the pretraining phase, we freeze the language model while training the entire framework end-to-end to optimize the weights of the vector encoders and the vector transformer (V-former). Such an optimization process enables effective grounding of the vector representation into an embedding that can be directly decoded by the LLMs. It is important to note that during this pretraining phase, we use only perception structured language labels and avoid training on tasks involve reasoning, such as action prediction (vehicle's control commands) and agent's attention prediction (where does the expert pay spatial attention to). This is because our focus at this stage is solely on representation training, and we aim to avoid prematurely integrating any reasoning components into the V-former.

The pretraining process was conducted using 100k question-answer pairs, which were collected from a simulator. Additionally, in each epoch, we sampled 200k vectors with uniformly distributed random values, which enhanced robust representation learning by comprehensively exploring the vector space and its associated semantic meanings. We employed the lanGen method, as described in Sections III-B, to automatically label the pseudo vector captioning data. During the pretraining phase, we optimize the vector encoder and V-former weights to transform the vector space into LLM-understandable language embeddings, by penalizing the error in the vector captioning results.

Through this approach, we are able to effectively incorporate object-level vector modality into pre-trained LLMs, which provide a good starting point for the finetuning stage.

*2) **Driving QA Finetuning**:* After the pre-training stage, we ingest the trainable LoRA module to the LLM, and optimize the weights Vector Encoder, Vector Former and LoRA module in an end-to-end fashion on the Driving QA data we collected in Section III-C.

In order to train the LLM-Driver to output accurate action for driving, we added certain action-triggering questions to the VQA dataset. These are questions that, when asked, require the agent to generate actions in specific format. For example, a question like "How are you going to drive in this case and why?" would require the agent to infer the action based on the vector input. We then use a simple language-action grounding strategy with regular expression matching to extract the action required to control the car in our simulator. To ensure the model pays sufficient attention to these important questions, we upsample the action-triggering questions with different expressions such as:

- "How are you going to drive in this situation?"
- "What actions are you taking?"
- "How are you driving in this situation?"
- "What are your planned actions for this situation?"

We refine the pre-existing model on 10,000 scenarios, each incorporating 16 driving QA entities, along with an additional vector captioning data and an action prediction QA pair. The outcome is a driving LLM agent that is capable of reasoning about its actions and any driving related questions given the object-level vectors of the current driving scenario.

IV. Evaluation and Results

This section aims to thoroughly assess our model's performance using a reserved evaluation set comprising 1,000 distinct driving scenarios. The vector captioning and vector QA data for these scenarios were labeled following the same methodologies discussed in Sections III-B and III-C.

A. Evaluation of Perception and Action Prediction

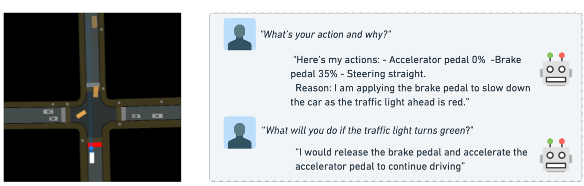

Fig. 4: Demonstration of the interaction with the pretrained LLM-Driver. Notably, the driving LLM agent possesses the ability to anticipate future actions when prompted.

To obtain the model's perception and action predictions, we prompt the model with the same vector captioning and the action-predicting triggering question added to the driving QA data, and then parse the language output using regular expressions to extract the numerical predictions.

We reported the results on the model trained using the two-stage approach (with pretraining), and the model training only on driving QA dataset (without pretraining). We also include a simple Behaviour Cloning (BC) model for comparison purposes, in order to assess the proficiency of leveraging the reasoning ability of pretrained LLMs in driving tasks (like action prediction). The BC model includes the identical Vector Encoder and V-former modules as the LLM agent model. However, it differs in the fact that it employs a transformer-based policy module in place of the LLM with adapters. The BC model also outputs actions with perception auxiliary tasks of agent and traffic light detection. To maintain an equitable comparison, we've calibrated the BC model to possess a similar count of trainable parameters as the LLM agent, totaling to approximately 25 million trainable parameters. The BC model was trained using the same 10k dataset that we used for the LLM Agent. This dataset only included perception and action data. We train the BC model for 5 epochs, same as the LLM Agent.

For the reported metrics, we calculate the Mean Absolute Error (MAE) for the predictions of the number of cars and pedestrians, denoted as E_{car} and E_{ped} respectively. Additionally, we measure the accuracy of traffic light detection as well as the mean absolute distance error in meters for traffic light distance prediction, represented as Acc_{TL} and D_{TL}. Furthermore, we compute the MAE for the normalized acceleration and brake pressure denoted as $E_{lon.}$, and normalized steering wheel angle denoted as $E_{lat.}$. Lastly, we report the weighted cross entropy loss for the token prediction on the evaluation set, indicated as L_{token}.

As can be seen in Table I, the results clearly demonstrate that the pretraining stage significantly enhances the model's both perception and action prediction capabilities. This suggests that the pretrained model exhibits a higher level of accuracy in perceiving and quantifying the number of cars and pedestrians in its environment. The pretrained model also shows a lower loss value, L_{token}, which indicates an improvement in the overall effectiveness of the model's token predictions.

Note that we filter the agents fall out of the 30m range from the ego vehicle, which need to be calculated through the x,y,z vector. This setting makes the “direct decoding” of the agent detection from vector much harder. For the simpler regression task (e.g. traffic light detection/distance), BC performs much better than LLMs.

For the action prediction task requiring in-depth reasoning, we found that LLM-based policies outperform the BC approach under the same amount of training data and trainable parameters. This indicates that LLMs serve as effective action predictors, harnessing knowledge acquired during the general pretraining phase (e.g., stopping at a red light, decelerating when vehicles or pedestrians are ahead) to inform decisions based on their grounded observations.

However, it's important to note the distinction in training methodologies: BC is trained using regression on perception and control outputs, while LLMs are trained via cross-entropy token loss and benefit from extra 16x pairs of driving question answering, which will reinforce the learning of perception and action prediction. Thus, the comparison might not be entirely equitable and should be taken as a point of reference.

TABLE I: The evaluation result of perception and action prediction

	agents count		*traffic light*		*action*		*loss*
	$E_{car} \downarrow$	$E_{ped} \downarrow$	$Acc_{TL} \uparrow$	$D_{TL} \downarrow$	$E_{lon.} \downarrow$	$E_{lat.} \downarrow$	$L_{token} \downarrow$
BC	0.934	1.459	**0.887**	**1.965**	0.181	0.094	n/a
LLM-Driver$_{\text{w/o pretrain}}$	0.187	1.683	0.696	8.116	0.093	0.015[a]	0.749
LLM-Driver$_{\text{w/ pretrain}}$	**0.149**	**0.435**	0.713	7.323	**0.067**	**0.015**[b]	**0.662**

[a] Exact value: 0.01497
[b] Exact value: 0.01495

B. Evaluation of Driving QA

To assess the quality of answers to open-ended questions about the driving environment, we use GPT-3.5 to grade our model's answers which is a recently emerging technique grading natural language responses [46] [47] [48]. This allows us to quickly and consistently evaluate our model's capabilities on questions without a fixed answer.

For evaluation we prompt GPT-3.5 with the language-based observation description used during dataset generation (section III-C), the question from the test set, and the answer of the model. GPT's task is to write a one-line assessment followed by a score from 0 to 10 (where 0 is worst and 10 is best) for each response, based on the given observation details. The final score of the model is the average over all question scores. Noticing that GPT evaluations can sometimes be overly lenient with answers that are incorrect but semantically close, as well as other issues with GPT-based grading reported by[48], we hand-scored 230 randomly sampled QA pairs for validation and obtained comparable results.

TABLE II: Grading of the Driving QA outputs

	GPT Grading ↑	Human Grading ↑
Constant answer "I don't know"	2.92	0.0
Randomly shuffled answers	3.88	0.26
LLM-Driver$_{\text{w/o pretrain}}$	7.48	6.63
LLM-Driver$_{\text{w pretrain}}$	**8.39**	**7.71**
Answers generated by GPT	9.47	9.24

Our results in Table II show that pre-training improves the grading score of the model by 9.1% and 10.8% over no pre-training from GPT grading and human grading. For reference, we also provide the score when running our evaluation procedure with artificially incorrect answers (by constantly answering "I don't know" or randomly shuffling all answers), as well as the "ground-truth" answer labels provided by GPT. We can see that our model, using only vectorized representations as input, can get a much higher score than constant answer or random answer.

V. Conclusion and Limitations

While our approach exhibits considerable novelty and potential, the preliminary results indicate that there's still progress to be made for the LLM to fully navigate in simulation. We are aware of possible discrepancies in open-loop vs closed-loop results [49], and our future work will be addressing the challenge of efficiently evaluating the model in a closed-loop system. This includes lengthy inference time of LLMs and the extensive steps needed to thoroughly test the system. Moreover, we anticipate the need to improve the precision of the direct driving commands produced by our baseline for it to operate effectively in closed-loop settings. Factors contributing to this include the intricacy of the task, potential enhancements to the model architecture, and the need for improving the scale and quality of our training dataset. We have observed that numeric inaccuracies and the early developmental stage of our system can lead to discrepancies in explanations and reasoning, preventing our ideas from being fully realized in closed-loop settings. These observations will guide our future research as we continue refining our approach

Nevertheless, our work establishes a foundation for future research in this direction. We believe that our proposed architecture, combined with the novel way for grounding a new modality into LLMs, data auto-labelling pipeline, and LLM-based evaluation pipeline, can serve as a starting point for researchers interested in exploring the integration of numeric vector modality with LLMs in the context of autonomous driving.

In terms of future research, improving the architecture to handle nuances in the numeric vector modality might be a promising direction. Our approach holds potential for application beyond simulated environments. Given sufficient real-world perception labels for pretraining and fine-tuning a VLM, our methodology could be adapted to real-world driving scenarios.

Overall, while our results are preliminary we believe our work is a significant step forward in the integration of vector modality with LLMs in the context of autonomous driving.

VI. Ethical Implications and Broader Impact

By introducing LLMs into autonomous driving we are inheriting their ethical implications [50]. We believe that those problems are going to be addressed by further research by the LLM community. As for autonomous driving. it's crucial to ensure that the system can handle all possible driving scenarios safely. By improving the interpretability of autonomous driving systems, we can help build trust in this technology, accelerating its adoption and ultimately leading to safer and more efficient transportation systems.

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